Vanguard

RAVENCLAW PRESENTATION

Digital Challenge
A/B Testing



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Ironhack Data Analytics · Project 2 · Mar – Jun 2017

PROJECT BACKGROUND

The goal of the project

The challenge

Vanguard wanted to know if a modern interface with in-context prompts helps clients complete their online process without getting stuck.

The experiment set-up

Clients were randomly split from March to June 2017: Control saw the traditional website, Test the new design.



Control: 23,532 clients (47%) **Test:** 26,968 clients (53%) Same 5 steps in both versions

PROJECT BACKGROUND

Hypotheses: three questions, one test each

MAIN QUESTION

Completion rate

Did more people finish?

Two-proportion z-test · plus the 5% business threshold

H0 (Null): Rate_new <= Rate_old
The new design is no better or is actually worse

H1 (Alternative): Rate_new > Rate_old
The new design gets more people to finish

SPEED QUESTION

Time per step

Is it faster?

One-sided t-test, per step

H0 (Null): Time_new >= Time_old
The new design takes the same amount of time or longer

H1 (Alternative): Time_new < Time_old
The new design helps users finish faster

FRICTION QUESTION

Backward clicks

Does it cause more errors?

Two-proportion z-test on actions

H0 (Null): Error_new = Error_old
The new design does not cause more errors than the old design.

H1 (Alternative): Error_new ≠ Error_old
The new design does cause more errors/frustration.

EXPLORATORY DATA ANALYSIS

Who uses Vanguard's online platform?

Client age and wealth

47

average age

$63k

median balance (mean $147k: a few very large accounts)

Customer tenure

12 yrs

average time as a client

11 yrs

median: most are long-standing clients

Behavioural activity

6

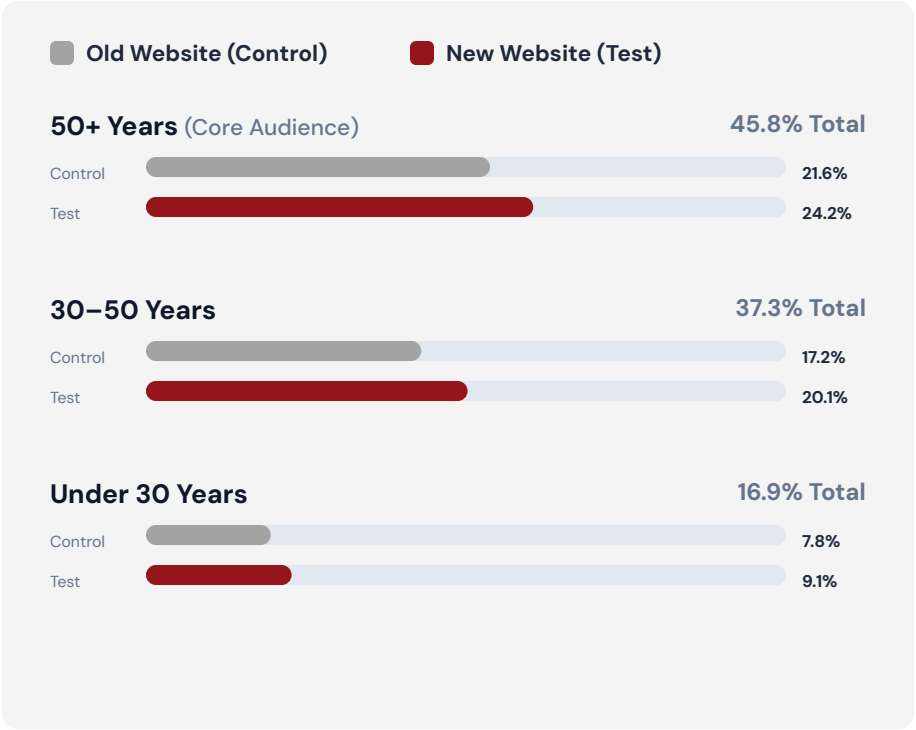
web logons in 6 months

3

support calls in 6 months

Source: client profiles of the 50,500 clients in the experiment

Age Group Distribution: Old vs. New Website



01

Perfect Age Proportions

Both website groups mirror each other precisely, confirming random assignment was completely unbiased.

02

Dominant Core Segment

Clients aged 50+ represent nearly half of all platform traffic, making them Vanguard's primary demographic.

03

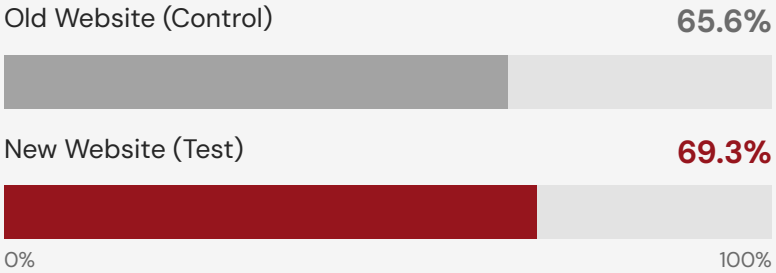
Validates Test Results

Because user profiles match evenly, completion rate differences are driven entirely by the new website design.

Old Website vs. New Website: Completion Rate Lift

Overall completion rate

Share of clients who reached the confirm page



+3.7 pts more clients complete the process on the new website

p < 0.001

Highly significant: H0 rejected, the 3.7-point lift is real.

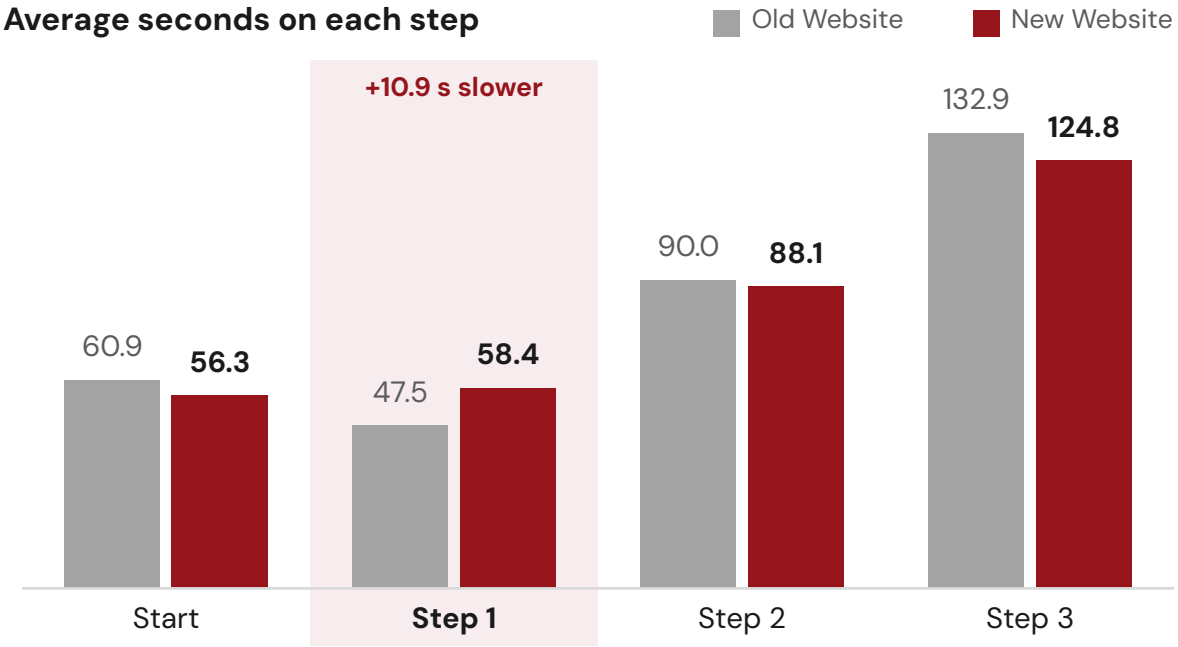
1. **Two-proportion z-test**
   Compared completion rates of 23,532 Control and 26,968 Test clients.
2. **Large sample**
   Over 50,000 clients give the power to detect small differences.
3. **Not chance**
   With no real difference, a lift this large would appear in under 0.1% of samples.

Source: experiment roster, client profiles and web logs, 15 Mar – 20 Jun 2017 (50,500 clients, 317,235 actions).

PERFORMANCE METRICS · TIME SPENT ON EACH STEP

Faster on three of four steps; step 1 takes 11 seconds longer

Average seconds on each step



+11 s

on step 1 with the new design (47.5 → 58.4 s)

- Reject H0 for start, step 2 and step 3: significantly faster
- Step 1: H0 not rejected (p ≈ 1.0)

Time on step = seconds until the next action in the same visit; pauses over 30 minutes and the confirm page excluded.

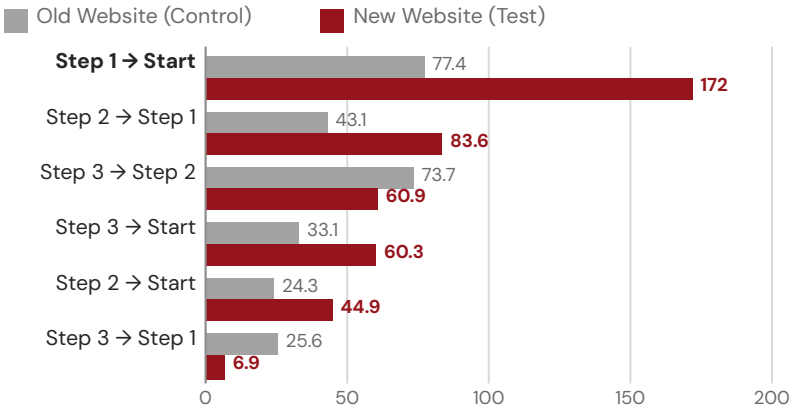
Error rate: more backward clicks, mostly from step 1

9.2% vs 6.8%
of actions go back a step (p < 0.001)

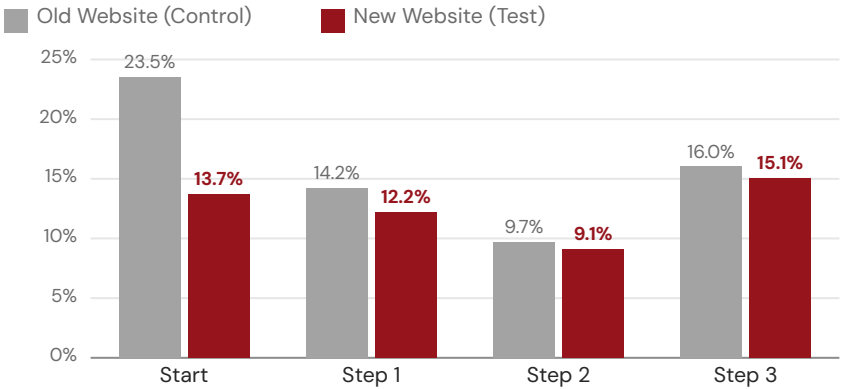
33% vs 26%
of clients make at least one error

43% vs 64%
finish, with vs without an error (new)

Backward moves per 1,000 visits



Visits that stop at each step



Clients aged 50+ are hit hardest: +9 pts more visits with an error on the new design, against +3 pts for under-30s.

Source: web logs of 69,205 visits, 15 Mar – 20 Jun 2017. "Stop" = the furthest step a visit reached before leaving.

Experiment evaluation: two of three KPIs favour the new design

KPI	Old website	New website	Test · p-value	Result
Completion rate	65.6%	69.3%	z-test · < 0.001	**Better: H0 rejected**
5% threshold	bar: +3.3 pts	+3.7 pts	z-test · 0.154	**Not proven**
Time: start, steps 2–3	61 / 90 / 133 s	56 / 88 / 125 s	t-test · < 0.05	**Better on 3 of 4 steps**
Time on step 1	47.5 s	58.4 s	t-test · ≈ 1.0	**Worse**
Error rate	6.7%	9.2%	z-test · < 0.001	**Worse: H0 rejected**

- **Design is reliable:** random split, same period, matching profiles.
- **Duration is short:** 80% joined in tax season and KPIs were still moving in June.

Explore the interactive dashboard (filters by age, gender, tenure and balance):
public.tableau.com · VanguardABtestdashboard

LESSONS LEARNED

Conclusion: a better process overall, with one clear weak point

What worked

More clients finish (+3.7 points).

Faster on start, step 2 and step 3.

A sound, randomised test design.

What to fix

Step 1: 11 s slower and 2× more returns to start.

Errors cut completion from 64% to 43%.

5% threshold not yet proven.

Next time

Run longer and outside tax season.

Track device and click events on step 1.

Agree the 5% definition and the value of a completion.

FINAL BUSINESS VERDICT

Go live, after fixing step 1

THANK YOU · TEAM RAVENCLAW · QUESTIONS?